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Normalized Differential Kerr Microscopy An Advanced Method for Magnetic Imaging

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Based on differential imaging techniques we developed a new variant of magneto-optical microscopy, which more perfectly than previous methods eliminates non-magnetic contributions in the magnetic image. Such contributions may stem from topographic or structural inhomogeneities in the magneto-optical saturation difference signal. We demonstrate the advantage of this new method various examples.

Ausgehend von der herkömmlichen Differenzbildtechnik entwickelten wir eine neue Variante der magnetooptischen Mikroskopie, welche nichtmagnetische Beiträge zum magnetischen Bild besser als bisherige Verfahren eliminiert. Derartige Einflüsse rühren von topographischen oder strukturellen Inhomogenitäten im magneto-optischen Sättigungs-Differenzsignal. An verschiedenen Beispielen demonstrieren wir die Vorteile der neuen Methode.

1. Introduction

The quality of magneto-optical domain observation was strongly enhanced during the last two decades by the introduction of digital image processing methods [1 to 5]. This is particularly important for the observation of planar magnetization patterns on soft magnetic materials, using the longitudinal or the transverse Kerr effect. For every setting of a Kerr microscope, the instrument is sensitive to surface-parallel magnetization components along a certain axis **a**, while magnetization components perpendicular to **a** are invisible [4]. Combining images with more than one sensitivity axis, the planar magnetization pattern can be evaluated quantitatively [4, 6 to 8]. The quantitative methods rely on calibration of the Kerr intensities by saturated images along at least four directions, which makes them cumbersome in practice.

The procedure demonstrated in this paper applies primarily to direct images for a single sensitivity direction **a**. A method is presented that uses only a minimum of additional information to improve the image. It eliminates a problem that is sometimes encountered: The total Kerr intensity difference between images for positive and negative saturation along the sensitivity axis – the "saturation difference image" – may be non-uniform if the sample itself is structurally non-uniform. The resulting non-uniformity of the magneto-optic saturation difference image will be superimposed on the domain image, making the domain image hard to interpret correctly. Normalizing the raw difference image by the saturation difference image yields an improved domain image,

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as will be demonstrated for various typical examples for which the traditional procedure leads to unsatisfactory results. Eventually, we will demonstrate that the new normalizing procedure may also be extended to a quantitative procedure.

2. Procedure

In standard magneto-optical difference methods a "non-magnetic" reference image is stored and subtracted from the "magnetic" image. We call the observed image intensity $I(m_a)$, where m_a is the magnetization component along the chosen sensitivity axis. As reference image we may use the image $I(-1)$ observed when the sample is saturated along the negative (dark) sensitivity direction. This leads to the following expression for the displayed domain image intensity P_m:

$$P_m = \gamma[I(m_a) - I(-1)] + I_0 \,. \tag{1}$$

Here $I(m_a)$ is the incoming image intensity as seen by the camera, which depends on the magnetization component m_a along the sensitivity axis, and γ and I_0 are parameters adjusted to map the incoming signal into the 8-bit display interval $(1 \leq P_m \leq 255)$ usually employed in image processing. This standard technique suppresses most of the non-magnetic background and leads under favorable circumstances to clear domain images even when the magnetic pattern is almost invisible in the incoming image.

The new method, which improves domain contrast even under unfavorable conditions, relies on storing, in addition, the saturation difference intensity $I(1) - I(-1)$, and normalizing the domain difference image by this difference intensity. From practical experience, the most important reasons for saturation difference contrasts are

– Inhomogeneities in the illumination that are enhanced in the digital difference mode.

– Non-planar surfaces. Already slight deviations from planarity create strong contrasts in the polarizing microscope, some of which remain visible after the image subtraction step.

– Different crystal orientations in polycrystalline materials. Even for cubic materials, which are optically and magneto-optically isotropic, "metallographic" contrasts often show up due to a submicroscopic surface texture that may have polarizing effects. Even with differential imaging techniques grains with strongly differing orientation can usually not be examined together in Kerr microscopy.

– Particularly strong effects are expected and found in heterogeneous substances and in optically anisotropic polycrystalline materials.

To eliminate these unwanted image contributions we use, in detail, the following operations: By applying saturating fields we record a magneto-optical *saturation image* P_s under identical conditions as before for the domain image P_m,

$$P_s = \gamma[I(1) - I(-1)] + I_0 \,. \tag{2}$$

From P_m and P_s we may calculate a normalized Kerr image according to

$$P_{nor} = \bar{\gamma}\,\frac{P_m - I_0}{P_s - I_0} + \bar{I}_0 \,, \tag{3}$$

where $\bar{\gamma}$ and $\bar{I}_0$ replace γ and I_0 to map the resulting ratio into the target 8-bit image range again. The *normalized magneto-optical image* P_{nor} can be treated in the same way as a traditionally obtained image. For linear imaging conditions the resulting normal-

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ized Kerr image should be a function of the selected magnetization component m_a only. The quotient is calculated from two differential images obtained by the established differential technique. If the regular differential imaging procedure is performed in a dedicated image processing unit, the resulting 8-bit images must be transferred to a computer. For regular points only the simple operation (3) has to be performed on these images. Some care has to be taken to deal with artifacts, resulting in particular from points for which the saturation image intensity $P_s - I_0$ vanishes. This will happen at non-magnetic inclusions or etch pits. Other irregular pixels may also be detected, such as points that display an anomalously high saturation intensity due to uncompensated lateral displacements between the two saturation images in (2). All these points may be assigned a neutral value in the final image (3). It may also be possible to eliminate a superimposed Faraday effect from the saturation image (2) if strong fields had to be applied, and if this Faraday effect (generated for example in the lens and in the immersion liquid) is a linear function of the applied field.

In practice, the saturation reference image $I(-1)$ in (1) is usually replaced by a "dynamic" reference image. In a dynamic reference image the domains are agitated in an alternating field along the sensitivity axis and the image is formed by averaging over the alternating pattern. The advantage of this method is that small ac fields usually suffice in this operation, so that displacements between reference and actual images are reduced and parasitical polar Kerr effect contributions are avoided [6]. The use of a dynamic reference image amounts in our procedure to replacing $I(-1)$ by $I(0)$ in (1), which means that only another normalizing parameter $\bar{\gamma}$ in (3) has to be chosen to reach the same goal.

Note that normalizing procedures similar to (3) were used in electron polarization methods of magnetic domain observation from the beginning [9, 10]. Because high-quality Kerr images can in many cases also be obtained without this additional step of image processing, this obvious option has up to now not been adopted in magneto-optics.

3. Verification and Demonstration

The new procedure is verified on some examples. In all cases the normalized Kerr technique eliminates non-magnetic influences in a better way than the usual technique, and improves the chances for a correct interpretation of the observed structures.

Metallographic contrasts on polycrystalline samples are particularly cumbersome for nickel-rich NiFe alloys (cf. [11, Fig. 5.44]) for two reasons: (i) Kerr contrast in these materials is rather weak compared to iron-rich alloys. (ii) The face-centred cubic crystals of NiFe alloys often contain twin boundaries that resemble in their appearance magnetic domains, leading to a wrong interpretation of conventional domain images as in Fig. 1c. The magneto-optical saturation image b) displays most clearly these straight-edged crystallites, while they virtually disappear in the normalized Kerr image d). Note that b) displays on the right edge what appears to be a row of foreign phase magnetic inclusions that do not show up in the direct image a). Magneto-optical saturation images may thus be useful in the metallographic analysis of all kinds of steels – in addition to their use in the present context.

A similar set of images is shown in Fig. 2 for a conventional silicon–iron transformer steel (cf. [11, Section 5.3.2]). Here the dark appearance of the strongly misoriented crystallite in the upper centre visible in the saturation image b) impairs the visibility of the fine supplementary domains in this crystal in conventional Kerr microscopy c). The normalized image displays uniform contrast for wide *and* fine domains over the whole field of view.

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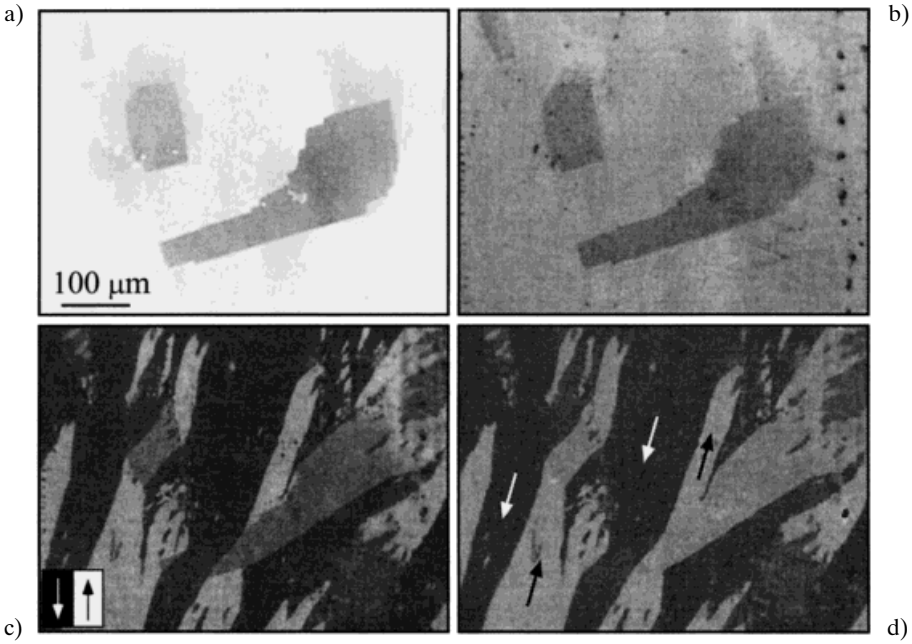

Fig. 1. Magneto-optical observation on a $Ni_{50}Fe_{50}$ polycrystalline sample. a) Shows the directly observed image. Due to different crystallographic orientations a local inhomogeneity of the Kerr amplitude occurs in the magneto-optical saturation image (2) that is shown in b). The conventional difference image (1) shown in c) suffers from superimposed grain contrast. The non-magnetic structures visible in b) also show up to some extent in c). In the normalized image d) these structural contrasts are largely eliminated, leaving a pure magnetic domain image

The flatness of the sample surface is generally an important requirement in Kerr microscopy. The natural surfaces of metallic glass samples (Fig. 3; cf. [11, Section 5.4]), but also electrolytically polished surfaces are usually less than ideal in this respect, so that direct domain observation at low magnification is virtually impossible in such samples (Fig. 3a). Conventional difference microscopy offers decisively better results c), but still displays some non-magnetic features. Note in particular the discretization artifacts visible in the saturation difference image b), which are connected with the large intensity variations visible in a). These artifacts – weak irregular dark lines – outline the low-intensity regions in a). They interfere with magnetic contrast especially in the lower right quadrant of c), but they are clearly suppressed in the normalized image d).

The new procedure is also useful in the investigation of transparent magnetic multilayers. The magnetic effect of different modification in the multilayer setup is preferably studied in "wedge" or "step" designs, in which, for example, the thickness of an interlayer is modified over the width of the sample. Such a case, in which the modification also has a strong optical effect, is shown in Fig. 4. Here a non-magnetic interlayer is inserted in the right half of the image. The effective "dilution" of the magnetic material, together with the change in reflectivity, leads to a reduced magneto-optical contrast for the part with the non-magnetic interlayer. Without the normalization procedure the domains in the two parts of the sample can hardly be compared.

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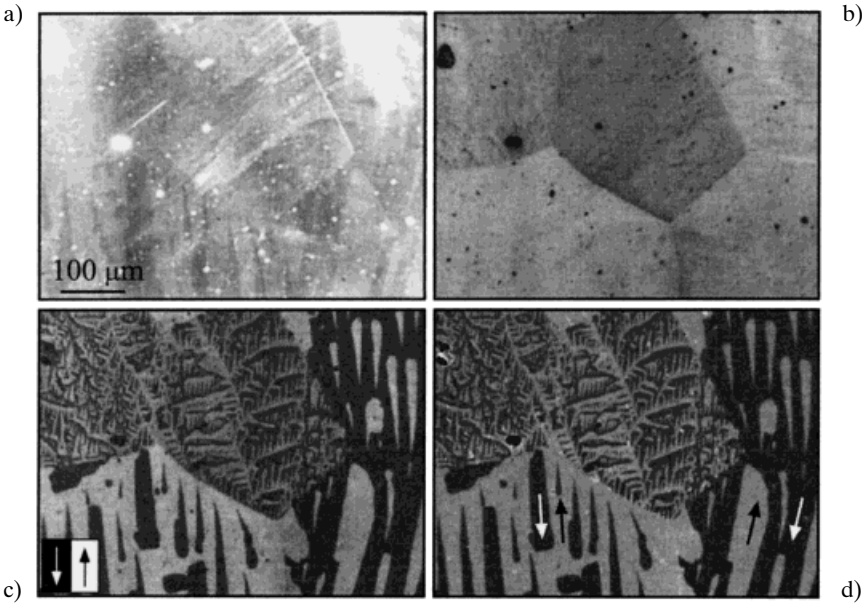

Fig. 2. Domain observation of Goss-textured silicon–iron transformer steel. The four images correspond to those in Fig. 1. The five grains have different crystallographic orientations, which can be identified from the observable domain patterns. In contrast to the regular difference image c), the normalized image d) displays a uniform domain contrast

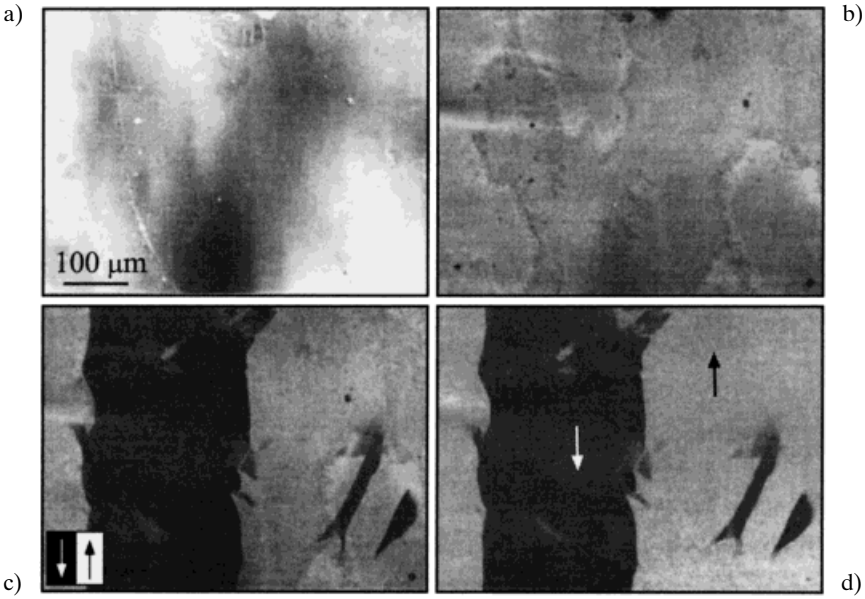

Fig. 3. Domain observation on a $(Co_{0.94}Fe_{0.06})_{79}Si_6B_{15}$ magnetostriction-free metallic glass. The four images are arranged in the same way as those in Fig. 1. Note the discretization contour in the saturation image b), which outlines the surface-profile-induced low-intensity zone in the direct image a). Such artifacts are suppressed in the normalized image d), in contrast to the regular difference image (c; particularly in the lower right corner)

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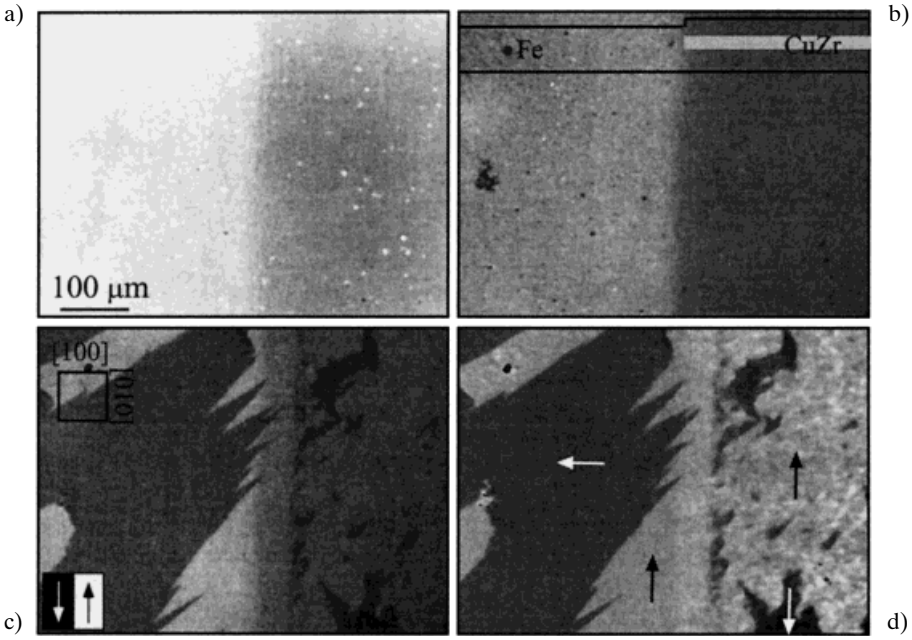

Fig. 4. Domain observation on a Fe(5 nm)/CuZr(4.4 nm)/Fe(5 nm)-sandwich structure (right side of the image). On the left side the CuZr-interlayer is omitted, leading to a Fe(10 nm) single layer. The optical effect of the copper–zirconium interlayer, which reduces the domain contrast in the conventional difference image b), is removed in the normalized image d) (Sample: C. Schmidt, D. Bürgler, University of Basel [12])

4. Quantitative Imaging

The introduced method can be extended to quantitative domain imaging, because the normalized images directly represent the magnetization component parallel to the plane of incidence. Two magnetic domain images P_m and two saturation images P_s with per-

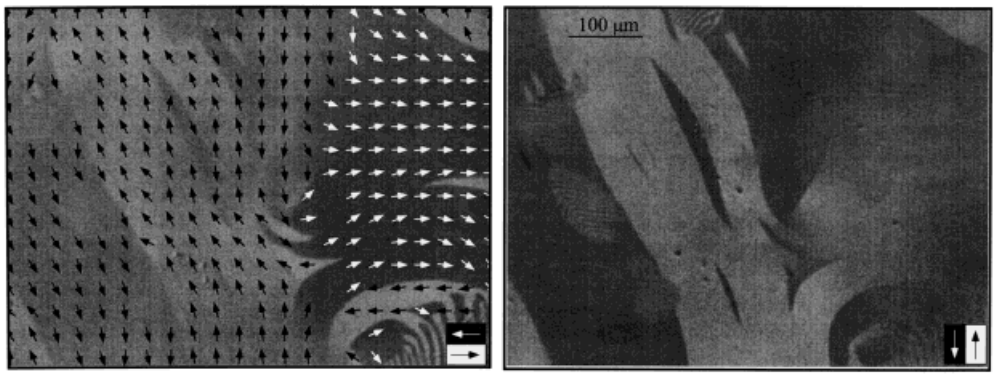

Fig. 5. Normalized quantitative images of an iron-rich metallic glass. Shown are the computed images for the x and they y magnetization components, respectively. In the left image the computed magnetization vector distribution is superimposed in suitable parts of the image

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pendicular sensitivity axes have to be stored and processed. Combining both normalized images, the angle of magnetization can be determined directly. As an example the case of a metallic glass is shown in Fig. 5. It should be mentioned that this procedure requires a well defined adjustment of the microscope.

5. Discussion

The presented method completes the well established differential techniques. The experimental realization is more complicated than the standard method because it needs an extra, off-line image processing step. The resulting domain images display the magnetic components with strongly reduced errors from lateral non-magnetic inhomogeneities. It might be possible, however, to incorporate the procedure into a real time image processing system, which is usually preferred in domain observation [3], because the needed saturation image P_s defined in (2) can be stored beforehand. Not more than one additional frame would have to be stored. The needed image processing operation (3) applies to 8-bit data only. It will therefore be possible to calculate the normalized image in real time (meaning at video frequencies) although a division operation is involved in (3). Alternatively, the operation (3) may be converted into ordinary difference operations by expansion, if the general background intensity is chosen large compared to the magneto-optical signal.

As a by-product, magneto-optical saturation images are obtained (parts b in Figs. 1 to 4). These images may be useful in the structural investigation of magnetic materials and alloys. They differ from the direct image (parts a in Figs. 1 to 4) in the polarization microscope, and as difference images, they suppress to some degree surface flaws and relief effects. Magneto-optical saturation images should be especially useful in detecting magnetic phases that differ in chemical composition or crystal structure and therefore in the strength of their magneto-optical material parameter.

6. Conclusions

The established digital difference technique of magneto-optical domain observation can be enhanced by taking into account the magneto-optical saturation image. In this way, the effects of inhomogeneities in the illumination, in composition and in crystal orientation can be eliminated. The presented method also provides a new variant of quantitative magneto-optical domain observation. Finally, magneto-optical saturation images may be useful as an additional tool in the metallography of steels and magnetic materials.

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